

F2. Guess the K-th Zero (Hard version)

time limit per test: 4 seconds
memory limit per test: 256 megabytes

This is an interactive problem.

This is a hard version of the problem. The difference from the easy version is that in the hard version $1 \leq t \leq \min(n, 10^4)$ and the total number of queries is limited to $6 \cdot 10^4$.

Polycarp is playing a computer game. In this game, an array consisting of zeros and ones is hidden. Polycarp wins if he guesses the position of the k -th zero from the left t times.

Polycarp can make no more than $6 \cdot 10^4$ requests totally of the following type:

- $? \ l \ r$ — find out the sum of all elements in positions from l to r ($1 \leq l \leq r \leq n$) inclusive.

To make the game more interesting, each guessed zero turns into one and the game continues on the changed array. More formally, if the position of the k -th zero was x , then after Polycarp guesses this position, the x -th element of the array will be replaced from 0 to 1.

Help Polycarp win the game.

Interaction

First, your program must read two integers n and t ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq t \leq \min(n, 10^4)$).

Then t lines follow, each of which contains one integer k ($1 \leq k \leq n$). It is guaranteed that at the moment of the request the array contains at least k zeros. In order to get the next value of k , you must output the answer for the previous value of k .

After that, you can make no more than $6 \cdot 10^4$ requests in total.

Use the following format to output the answer (it is not a request, it doesn't count in $6 \cdot 10^4$):

- $! \ x$ — position of the k -th zero.

Positions in the array are numbered from left to right from 1 to n inclusive.

After printing t answers, your program should exit immediately.

In this task, the interactor is **not adaptive**. This means that within the same test, the hidden array and the queries **do not change**.

In case of an incorrect query, -1 will be displayed. When this value is received, your program must immediately exit normally (for example, by calling `exit(0)`), otherwise, the testing system may issue an arbitrary verdict.

If the number of requests is exceeded, the verdict *wrong answer* will be displayed.

Your solution may get the verdict *Idleness limit exceeded* if you don't print anything or forget to flush the output buffer.

To flush the output buffer, you need to do the following immediately after the query output and the end-of-line character:

- `fflush(stdout)` or `cout.flush()` in C++;
- `System.out.flush()` in Java;
- `flush(output)` in Pascal;
- `stdout.flush()` in Python;
- see the documentation for other languages.

Hacks

Use the following format for hacks:

On the first line print the string s ($1 \leq |s| \leq 2 \cdot 10^5$), consisting of zeros and ones, and an integer t ($1 \leq t \leq \min(|s|, 10^4)$) — hidden array and number of requests, respectively. In the next t lines output the number k ($1 \leq k \leq |s|$).

The hacked solution will not have direct access to the hidden array.

Codeforces Round 719 (Div. 3)

Finished

Practice

Virtual participation

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Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Last submissions

Submission	Time	Verdict
361337412	Feb/03/2026 21:46	Accepted
361337083	Feb/03/2026 21:44	Wrong answer on test 4
361335755	Feb/03/2026 21:33	Wrong answer on test 21

Problem tags

binary search constructive algorithms data structures interactive *2200

No tag edit access

Contest materials

Announcement (en) Editorial (en)

Example

input	Copy
6 2 2 2 1 1 0 1 0	
output	Copy
? 4 6 ? 1 1 ? 1 2 ? 5 5 ! 5 ? 2 2 ! 2	

Note

In the first test, the array [1, 0, 1, 1, 0, 1] is hidden. After answering the query $k = 2$, the array changed to [1, 0, 1, 1, 1, 1].

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